



Olist E-Commerce Analysis, Portfolio Presentation

By Jayhahn Booker

Project Overview / Business Problem

Project Overview:

- This project presents a comprehensive analysis of the Brazilian Olist E-Commerce dataset using SQL, Python and Tableau
- I conducted analysis of over 99,000 orders that revealed:
 - Customer Retention: 96.88% only make one purchase over the course of 2+ years
 - Customer Satisfaction: Office furniture 21% one star review rate
 - Seasonality / Revenue Concentration: 97% revenue spike in November and Top 1% sellers account for 25.67% of overall business revenue

Tools Used

SQL (SQLite)

- Multi table joins across up to 4 tables
- Aggregations, GROUP BY, and time based analysis using STRFTIME
- Answered 15 business questions across 4 domains

Python (Pandas, NumPy)

- Data cleaning and datetime conversion
- Statistical analysis, distributions, outliers, percentiles
- Deepened 5 key findings beyond SQL's surface level answers

Tableau

- Built 4 visualizations and an executive dashboard
- Geographic, time series, and comparative analysis
- Published interactive dashboard for stakeholder use

Key Finding 1: Customer Retention Crisis

- 96.88% of customers do not make a second purchase
- Of the 3.12% that do make a second purchase only 7% of them made an additional third purchase
- These findings suggest that customers either have bad experiences with this business or are not given enough incentive to return.

Key Finding 2: Poor Customer Satisfaction (Late Deliveries)

- Late deliveries have an average review score of 2.57 while on time deliveries have an average review score of 4.21
- The average late delivery is 8.87 days late on average
- 7.87% of all deliveries were delivered past their estimated arrival date
- Revenue at risk analysis revealed that R\$1,357,314.96 which is 8.44% of total revenue, as being associated with late orders and dissatisfied customers

Key Finding 3: Seasonality and Revenue Concentration

Seasonality:

- November had a **97%** spike in revenue in 2017
- Suggests large black friday spike which is a popular day in Brazil
- **Solution:** Close the gap between slower months and top performing months (November) and slower months (February) for more consistent revenue.

Revenue Concentration:

- **Top 1%** of sellers (**30 individual sellers**) generate **25.67%** of total platform revenue
- **Top 10%** sellers account for **67.49%** of total business revenue
- **Bottom 50%** of sellers (**1,547 sellers**) account for only contribute only **3.22%** of business revenue
- **Solution:** Help bottom sellers generate more sales

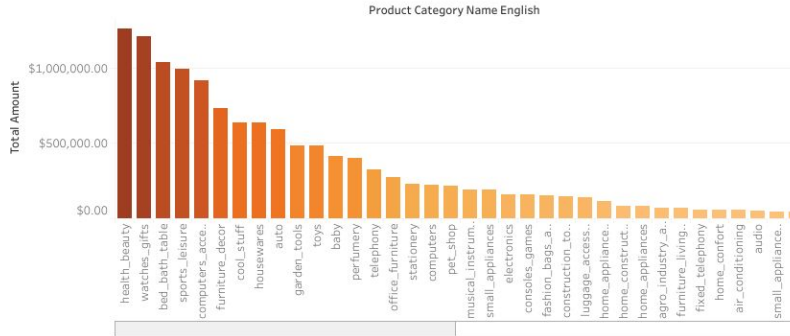
Olist's revenue is highly dependent on specific time periods and specific sellers, both represent areas of strategic vulnerability and opportunity.

Strategic Recommendations

- Recommendation 1: **Launch a customer loyalty program** targeting the second purchase retention cliff
- Recommendation 2: **Implement a seller fulfillment scorecard** to reduce late deliveries and protect satisfaction
- Recommendation 3: **Scale operations seasonally** ahead of November demand, promote during slow months
- Recommendation 4: **Create a top seller retention program** for the highest revenue sellers
- Recommendation 5: **Adopt category level seller evaluation** to ensure fair, accurate performance metrics

Tableau Dashboard

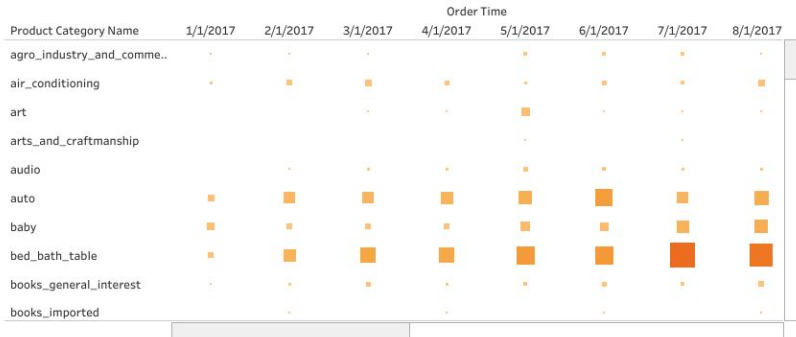
Revenue By Product Category



Average Delivery Time By State (Days)



Monthly Revenue By Category



Top 10 Sellers — Revenue Vs Customer Satisfaction



Tableau Dashboard Link:

https://public.tableau.com/views/OlistEcommerceAnalysis_17807796251430/Dashboard1_1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

What This Project Demonstrates

- Translating business questions into data queries
- Identifying patterns that connect operational issues to revenue impact
- Communicating findings to both technical and non technical audiences
- Structuring open ended analysis into a clear strategic narrative
- Going beyond surface level metrics to find root causes

Thank You!

Email: jayhahnb@gmail.com

LinkedIn: www.linkedin.com/in/jayhahn-booker-433a19403

Github: <https://github.com/jayhahn-booker/olist-ecommerce-analysis/tree/main>